Week 1 - Introduction to Data Analysis

Concepts:

Understanding the role of a data analyst, Data analysis process, Data visualization basics

Project:

Create a personal vision board

Outcome:

Understand the role and process of a data analyst

Week 2 - Data Types and Sources

Concepts:

Types of data (quantitative, qualitative), Data sources (internal, external), Data quality and integrity

Project:

Research and document data sources

Outcome:

Understand different types of data and sources

Week 3 - Data Preprocessing

Concepts:

Data cleaning and handling, Data transformation and normalization, Data aggregation and grouping

Project:

Practice data preprocessing on a sample dataset

Outcome:

Understand data preprocessing techniques

Week 4 - Data Visualization

Concepts:

Types of data visualization (charts, tables, maps), Data visualization tools (Tableau, Power BI, D3.js),

Project:

Create a data visualization dashboard

Outcome:

Understand data visualization techniques and tools

Week 5 - Descriptive Statistics

Concepts:

Measures of central tendency (mean, median, mode), Measures of variability (range, variance, standa

Project:

Calculate descriptive statistics on a sample dataset

Outcome:

Understand descriptive statistics concepts

Week 6 - Inferential Statistics

Concepts:

Hypothesis testing (null and alternative hypotheses), Confidence intervals and margin of error, P-valu

Project:

Perform hypothesis testing on a sample dataset

Outcome:

Understand inferential statistics concepts

Week 7 - Regression Analysis

Concepts:

Simple linear regression, Multiple linear regression, Regression assumptions and diagnostics

Project:

Perform regression analysis on a sample dataset

Outcome:

Understand regression analysis concepts

Week 8 - Time Series Analysis

Concepts:

Time series data and visualization, Time series decomposition and forecasting, ARIMA and exponenti

Project:

Perform time series analysis on a sample dataset

Outcome:

Understand time series analysis concepts

Week 9 - Machine Learning Fundamentals

Concepts:

Supervised and unsupervised learning, Regression and classification algorithms, Model evaluation me

Project:

Implement a machine learning model on a sample dataset

Outcome:

Understand machine learning fundamentals

Week 10 - Data Storytelling

Concepts:

Data storytelling principles and best practices, Creating engaging and informative reports, Communica

Project:

Create a data-driven story on a sample dataset

Outcome:

Understand data storytelling principles and best practices

Week 11 - Data Ethics and Governance

Concepts:

Data privacy and security, Data governance and compliance, Ethical considerations in data analysis

Project:

Develop a data governance plan for a sample organization

Outcome:

Understand data ethics and governance principles

Week 12 - Career Development and Portfolio Building

Concepts:

Building a professional portfolio, Networking and career development strategies, Staying up-to-date w

Project:

Create a professional portfolio and career development plan

Outcome:

Understand career development and portfolio building strategies

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